To install Jenkins on a Google Compute Engine instance, you can follow these general steps:

1. **Create a Google Compute Engine Instance:**

   - Go to the Google Cloud Console: https://console.cloud.google.com/

   - Create a new Compute Engine instance. Choose a suitable machine type and ensure it has enough resources for Jenkins.

2. **Connect to Your Instance:**

   - Use SSH to connect to your Compute Engine instance. You can do this directly from the Google Cloud Console or using a terminal.

3. **Install Java:**

   - Jenkins requires Java. Install OpenJDK or Oracle JDK. For example, on Ubuntu, you can install OpenJDK with:

     bash

     sudo apt update

     sudo apt install default-jdk

4. **Add Jenkins Repository:**

   - Add the Jenkins repository to your package manager to ensure you get the latest stable version.

     bash

     wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo apt-key add -

     sudo sh -c 'echo deb http://pkg.jenkins.io/debian-stable binary/ > /etc/apt/sources.list.d/jenkins.list'

5. **Install Jenkins:**

   - Update your package index and install Jenkins.

     bash

     sudo apt update

```
sudo apt install jenkins
```

6. **Start Jenkins:**

   - Start the Jenkins service.

```bash
sudo systemctl start jenkins
```

7. **Access Jenkins Web Interface:**

   - Jenkins runs on port 8080 by default. Access Jenkins using your instance's external IP address followed by `:8080` in a web browser:

```
http://your-instance-ip:8080
```

   - Follow the on-screen instructions to complete the setup. You will need to retrieve the initial administrator password from the Jenkins logs.

8. **Configure Jenkins:**

   - Once Jenkins is running, configure it according to your requirements. Install necessary plugins, set up security, and configure agents as needed.

9. **Enable Jenkins Service (Optional):**

   - If you want Jenkins to start automatically on boot, enable the service:

```bash
sudo systemctl enable jenkins
```

10. **Secure Jenkins:**

   - Jenkins initial setup wizard will guide you through creating an admin user and setting security options. Follow these steps to secure your Jenkins instance.

That's it! You should now have Jenkins installed and running on your Google Compute Engine instance.

--

To run a specific version of Jenkins on Google Compute Engine, you typically follow the installation steps as outlined previously. However, if you need to install a specific version of Jenkins instead of the latest stable version, you can modify the steps slightly to specify the version during installation. Here's how you can do it:

1. **Create a Google Compute Engine Instance:**

   - Ensure you have a Compute Engine instance ready with sufficient resources.

2. **Connect to Your Instance:**

   - Use SSH to connect to your Compute Engine instance.

3. **Install Java:**

   - Jenkins requires Java. Install the appropriate version of OpenJDK or Oracle JDK.

4. **Add Jenkins Repository:**

   - Add the Jenkins repository to your package manager. Modify the repository URL to specify the version you want to install. For example, to install Jenkins 2.303.1, use:

     bash

     wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo apt-key add -

     sudo sh -c 'echo deb http://pkg.jenkins.io/debian binary/ > /etc/apt/sources.list.d/jenkins.list'

     Replace `binary/` with the appropriate version directory, e.g., `binary/2.303/`.

5. **Install Jenkins:**

   - Update your package index and install Jenkins. Specify the version explicitly:

     bash

     sudo apt update

     sudo apt install jenkins=2.303.1

Replace `2.303.1` with the version you want to install.

6. **Start Jenkins:**

- Start the Jenkins service:

bash

sudo systemctl start jenkins

7. **Access Jenkins Web Interface:**

- Jenkins runs on port 8080 by default. Access Jenkins using your instance's external IP address followed by `:8080` in a web browser:

http://your-instance-ip:8080

8. **Configure Jenkins:**

- Follow the on-screen instructions to complete the Jenkins setup. Retrieve the initial administrator password from the Jenkins logs as needed.

9. **Enable Jenkins Service (Optional):**

- Enable Jenkins to start automatically on boot:

bash

sudo systemctl enable jenkins

10. **Secure Jenkins:**

- Complete the Jenkins initial setup wizard to create an admin user and configure security settings.

This approach allows you to install a specific version of Jenkins on your Google Compute Engine instance by specifying the version during the installation process. Adjust the version numbers and repository URLs as necessary based on the Jenkins release you intend to use.